Mobile Banking Tech Stack Experience Sharing

A Trusted Partner for Digital Transformation

iSoftStone Team

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Future Vision

To build a fully integrated, real-time, and intelligent customer experience hub, the key measures include:

Curated Customer Experience	**Active real-time customer interaction**	**Seamless omnichannel experience** (Omni-channel)	**360° Customer Insights** (Real-time)	**Hyper-personalization**
Real-time Detection System of Fraud and Deception	**Single View of Customer**	**Unified Real-time Profile**	**cross system customer data integration**	**Customer Data Platform (CDP) Construction**
intelligent recommendation engine (Next Best Offer)	**real time analysis of customer behavior** Identify the optimal touchpoint timing	**Event-driven architecture (EDA) implementation**	**AI-enhanced development and testing capabilities**	**Modularized Architecture**

Current Situation Analysis and Challenges

Current state of technology architecture

Deployment environment: Private OP environment, not fully containerized

Elasticity issue: Traffic peak throttling, resource scaling is limited

Architecture Bottleneck: High Service Coupling Degree and Lack of Hierarchical Guarantee Mechanism

Database architecture hybrid: Oracle and MySQL/MariaDB are used in combination, lacking unified planning

Third-party dependency management: Third-party services (e.g., VKey) exist. The connection method must be standardized.

Potential Limitation in business scenarios

Customer experience is limited: system response is delayed during peak hours, transaction processing takes longer, and feature updates are slow

Data and Intelligence Capability Not Fully Utilized: Insufficient Real-time Interaction and Personalization, Lack of User Behavior-Based Personalized Recommendations and Services

Weak channel synergy: Online and offline channels are not fully integrated, resulting in inconsistent user experiences.

Low efficiency of operation and development: Lack of automation and intelligent support

Future Vision and Core Objectives

Future Vision: In alignment with UOB ‘s strategic vision to "establish a fully integrated, real-time, and intelligent customer experience hub," we will develop a customer-centric, technology-driven evolution plan to transform mobile banking from a transactional tool into a smart financial partner.

modernization of architecture

Decouple existing monolithic applications to build an elastic and highly available cloud-native microservices architecture

Experience intelligence

Integrate AI capabilities to enable hyper-personalized, proactive, and seamless customer interactions

Data-driven operations

Establish unified customer view and real-time analytics to drive precise decision-making

Agile delivery

Build an efficient DevOps system to support rapid business iteration and innovation

The Basic of Technology Stack-Cloud Native Microservice Platform Guided by DDD

Design Methodology

Core: domain driven design

(Domain-Driven Design)

Practice: Through the Event Storm

Event Storming Workshop with Business and Technical Experts
Identify bounded context

Technology Foundation

Containerization

Docker & Kubernetes Implement Elastic Scaling and Service Self-healing

Development Framework

Middle-Platform Components

Building Enterprise Microservice Component Library Based on Spring Boot

A Market Proven Mobile Application Architecture

99.99%	Billion Users	14 years	100%	Safety	Compatible	Gartner
High availability and stability	User daily activity verification	Stable support for 11.11 flash sales	Improved development efficiency	Built-in RSA/ECC and other encryption algorithms	Mainstream software and hardware	Gartner 2022 " Multi-Experience Development Platform Market Guide "

Solution

Mobile Development solution	Mini Program/ Mini GameSolution	User Growth / Experience Solution	Precision Marketing Solution	Digital Human /intelligent Customer Service Solutions
E-wallet Super App Solution	Smart Government / city superApp Solution	Smart Transportation solution	Mobile Banking Solution	Digital Banking Solution

Scenario Middle Platform Abilityy

Mini Programs / MiniGames

Mini Program Decelopment Kit	Mini Program Low Code
Mini Program Test	Mini Program Release
Mini Program Analysis	Mini Program Monitoring
Mini Program Market	Mini Program Store
Mini Program Open Platform	Mini Program Multi terminal operation

Data Analysis Operations &Business

Mobile Analytics	Dynamic Cards
Performance Monitoring APM	Smart Layout
Visualization of buried points	Network detection MTR
Intelligent Construction Lowcode	Network Monitoring
Crowd Operation	Intelligent Scheduling MDC

Marketing & ContentOperations

Search and push integration	Smart delivery MCDP
User Experience Platform	AB Testing
LBS	Terminal Intelligence
Application Center	Mars Animation
Precision Marketing	Content Platform

AI& Multimedia

AI Smart Assistant	AI Development Assistant
AI Intelligent Customer Service	Audio and Video
Live Video	OCR
Instant Message	Insurance quality inspection
Intelligent Dual Recording	Video Business Hall

R&D Efficiency & safety

Cloud IDE	D2C (Design to Cide)
R&D Collaboration	Mobile Testing
Safety Testing	Security Hardening
Privacy Compliance	Mobile Threat Awareness
Device fingerprint	Disaster Recovery
Real person Authentication	HUAWEI Harmony

InnovativeB usiness

- Digital Human
- IOT
- AR
- VR
- AI

Base Component

Framework

Native	H5 Container
Mini Program	Cube

UI

AntUI	Mini Prograrm UI
AntDesign	

SDK

Equipment identification	Version Upgrade	Scan the QR Code	Log point	Social Sharing	Webview Kernel
Performance Monitoring	Unified Storage	Hotfixes	Wireless Bodyguard	Office Package	...

MGS Gateway Service	MDS Publishing Service	MPS Messace Push	MSS Synchronization Service	MAS Mobile Analytics

Cloud Base

On-Permise , Alibaba Cloud , Aws ...

An Evolution Roadmap for Mobile Application Platform

UOB iSoftStone Digital

Build product Quickly

Efficiently Build high-performance Product to retain users.

Basic Service Components
- Mobile Gateway
- Publish Service
- MAS
- MSS
- Message Push

Client Components
- Native
- H5
- Hybrid
- Mini Program

R&D Devops
- Mobile Test Platform
- Mobile Secrity
- Collaboration Platform

Make the product right

Quality technologies to improve experience for users and Security technology to protect users.

Application Performance Management
- Real-time Config
- Remote Diagnostics
- Performance and Exception Monitoring

Customer Experience Management
- A/B Testing
- Operational Analysis
- In-App Survey
- APM

Security
- Threat Detection
- Security hardening
- Unique ID

Generate healthy and sustainable growth

Promotion and Engagement technologies to acquire, activate, and retain exponentially more users.

Acquisition
- User Experience Platform
- Spread

Activation and Retention
- Content Distribution Platform
- Message Center
- App Center
- Dynamic Card
- Feeds

Upgrade to Super App

Open Platform technologies to create ecosystem to allow 3rd party developers to create more scenarios for users.

Mini-Program Platform
- IDE
- Developer Portal
- AppContainer
- Global Mini Program Standard

Interactive Platform
- 3rd party ecosystem
- Leaderboard
- Gamification

Evolution Path of Technology Architecture: Transition from Monolithic Architecture to Modular Microservice Architecture

Architectural evolution pattern selection

evolutionary pattern	applicable scene	Recommended degree of adoption	implementation complexity
Strangler Pattern	Gradually replace the old system functions, and the new and old systems will coexist	★★★★★	Middle
Demolisher Pattern	Rewrite the system completely and replace it all at once	★★☆☆☆	High
Renovator Pattern	Optimize and improve the original system	★★★☆☆	Low

Recommended Evolution Strategy: Stifle Mode + Modular Transformation

Progressive migration: Gradually migrate functions from monolithic systems to microservices to reduce risks

Split functional modules: Split services by business functional modules guided by Domain-Driven Design (DDD)

Service registration and discovery: Introduce Consul or Nacos as the service registry

Unified API Gateway Entry: Build a unified API gateway to implement routing, rate limiting, and circuit breaking

Containerized Deployment: Implementing Service Containerization and Elastic Scaling Based on Kubernetes

Evolution Path of Technology Architecture-Infrastructure and Elastic Capability Upgrade (Level 1)

First Stage: Basic Platform Build

•Complete Kubernetes cluster deployment and optimization

•Implement containerization transformation for core applications

•Set up a CI/CD pipeline for automated deployment

•Introduce a service registration center (Consul/Nacos)

Phase 1 Phase 2: Microservice Governance

•Complete the first batch of microservice decomposition (by functional module)

•Deploy the API gateway to enable unified access

•Set up a monitoring and alert system (Prometheus + Grafana)

•Enable centralized log collection and analysis

Phase 1, Stage 3: Data Capacity Building

•Build a real-time data pipeline (Kafka+Flink)

•Create a basic version of the Customer Data Platform (CDP)

•Implement Single View of Customer

•Establish a data quality monitoring system

Phase 1, Stage 4: Introduction of Intelligent Capabilities

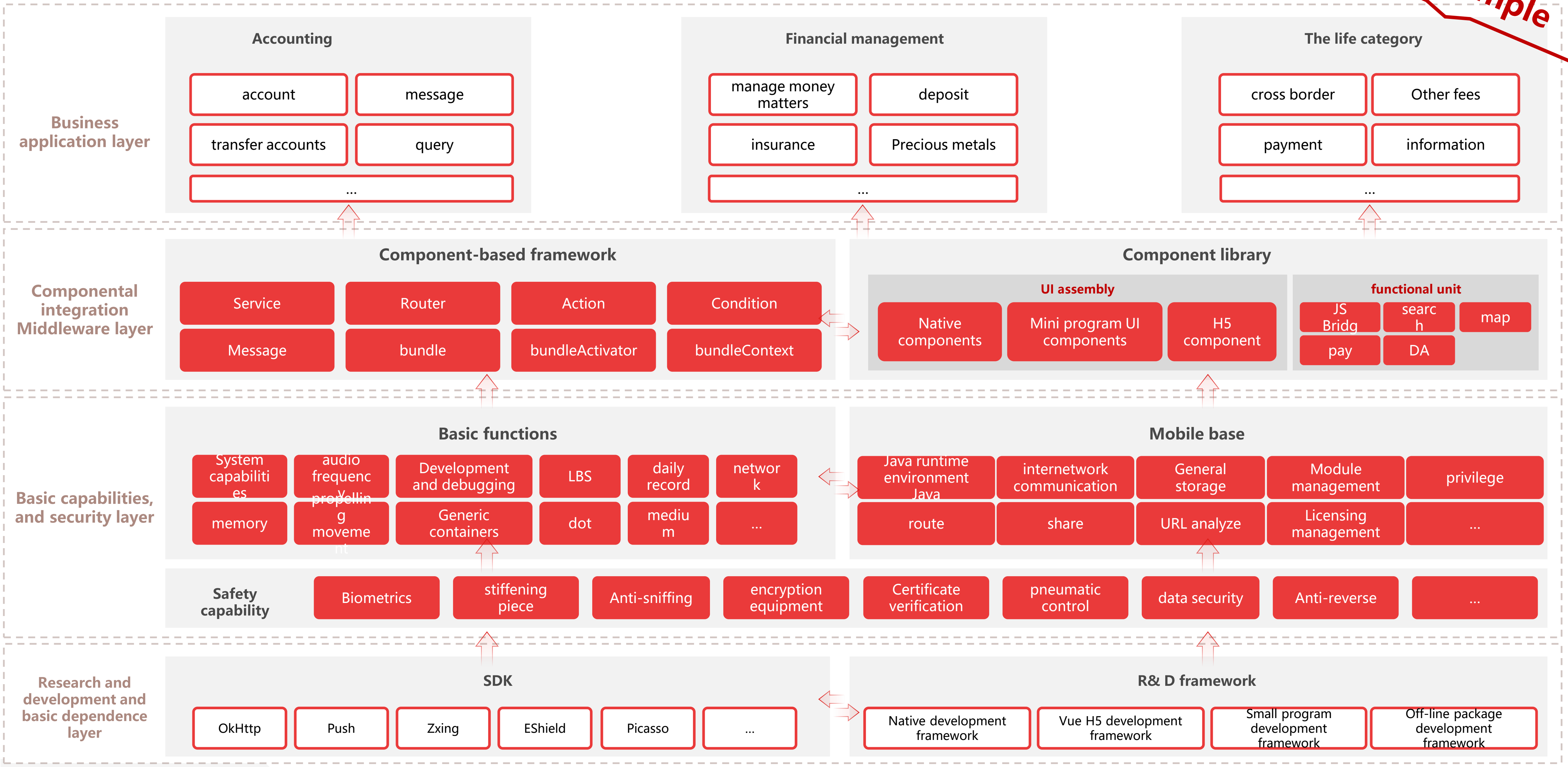
•Deploy rule engines to enable basic real-time risk control

•Introduce AI code assistants to improve development efficiency

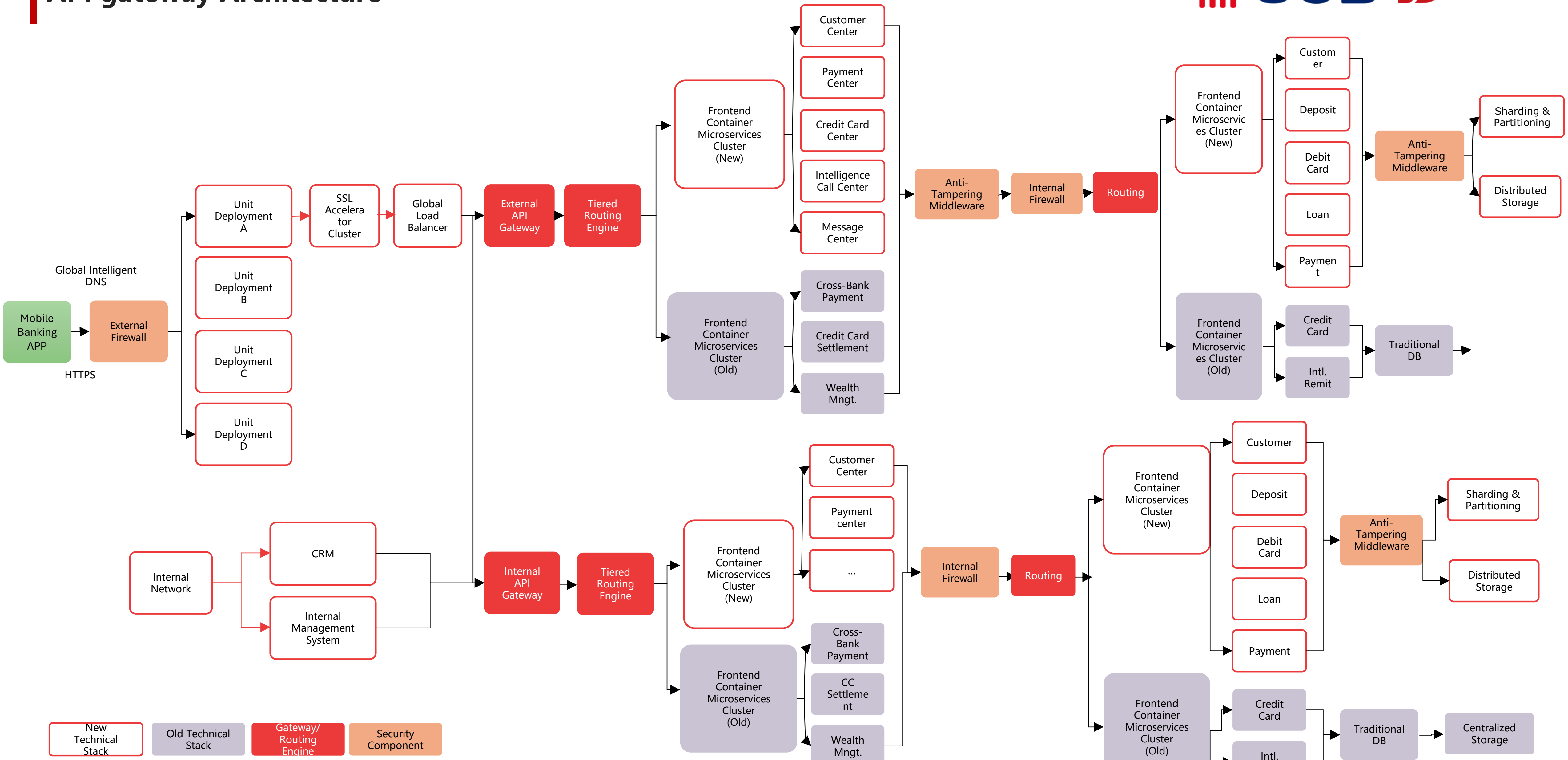
•Building the Basic Function of Intelligent Testing Platform

•Complete the unified entry point transformation for omnichannel experience

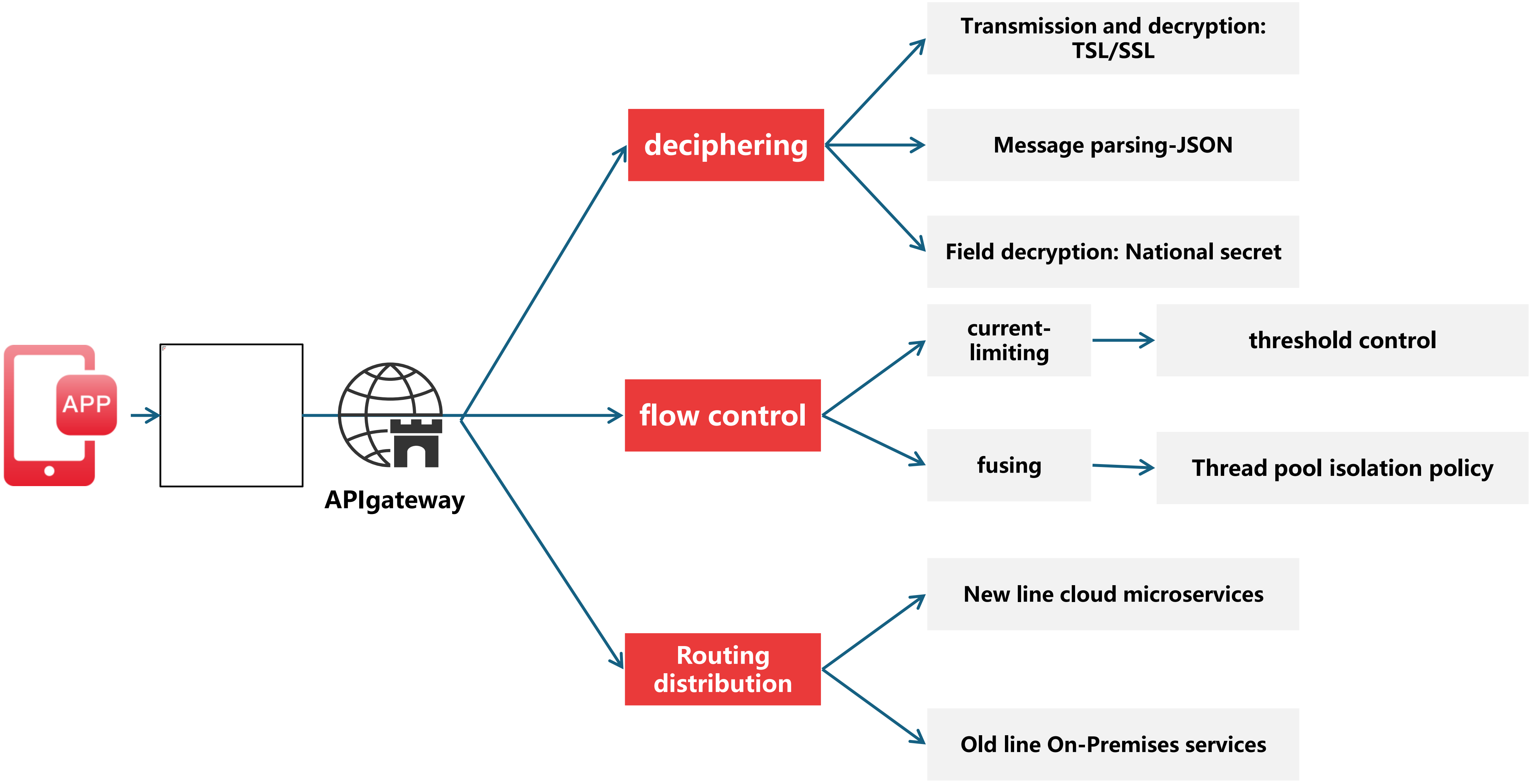
Front-end and Back-end Technology Architecture



API gateway Architecture



API gateway design diagram



Microservice design and decomposition scheme

Microservice split scheme

Split the premise

- There is a continuous integration platform to ensure continuous splitting and functional consistency
- The access layer API is managed by the API gateway to ensure smooth decomposition
- Database design avoids a large number of joint queries and reduces the coupling degree between the database and the table
- Application statelessness to ensure decomposability

Timing of the split	There are a lot of conflicts in the submitted code	Function modules are tightly coupled and can not be launched independently	
	Horizontal scaling process is complex ; primary and secondary businesses are tightly coupled	Meltdown downgrade depends on conditional statements (if-else)	
Split method	Engineering code standardization	Establish microservice engineering and progressive migration replacement	
	Standardization of test folders	Track and monitor to ensure a smooth transition	
Splitting Rules	Horizontal split	No more than three layers can be split	
	One-way call and prohibit circular call	Loop calls are converted to parallel calls or asynchronous calls	
	Interface data definition is strictly prohibited from being embedded or transmitted	Standardized project name	The interface implementation is idempotent

Core microservice split diagram

deposit	provide a loan	Payment clearing
credit card	cash administration	Money transactions
configuration service		
communication services		
data service		
dictionary	cache	storage

The split strategy can be considered in terms of functional and non-functional dimensions:

★ The functional dimension is to clearly define the boundaries of the business

The main design method adopted is DDD. The strategic design of DDD will establish the domain model, which can guide the decomposition of microservices.

★ There are six main non-functional dimensions

Scalability, reusability, high performance, high availability, security, and heterogeneity

The microservice split proposal is based on the gradual refactoring of the killer pattern, avoiding drastic changes, pursuing a long stream, continuous iteration, gradual migration, and smooth implementation

Microservice Architecture Design-Service Splitting Strategy Based on Functional Modules

Principles and Strategies of Microservice Splitting

Split Policy

- **By business function module:** Account services, Transfer services, Wealth management services, Credit services, etc.
- **High cohesion, low coupling:** Ensure services have closely related internal functions and minimize inter-service dependencies
- **Independent deployment and scaling:** Each service can be deployed, scaled, and upgraded independently
- **Data autonomy principle:** Each microservice maintains its own database to avoid shared databases

Technology stack selection

- **Development Framework:** Spring Boot + Spring Cloud
- **Service registration and discovery:** Consul/Nacos
- **API Gateway:** Spring Cloud Gateway
- **Configuration Center:** Apollo/Nacos Config
- **Fault tolerance and rate limiting:** Sentinel
- **Distributed Transactions:** Seata

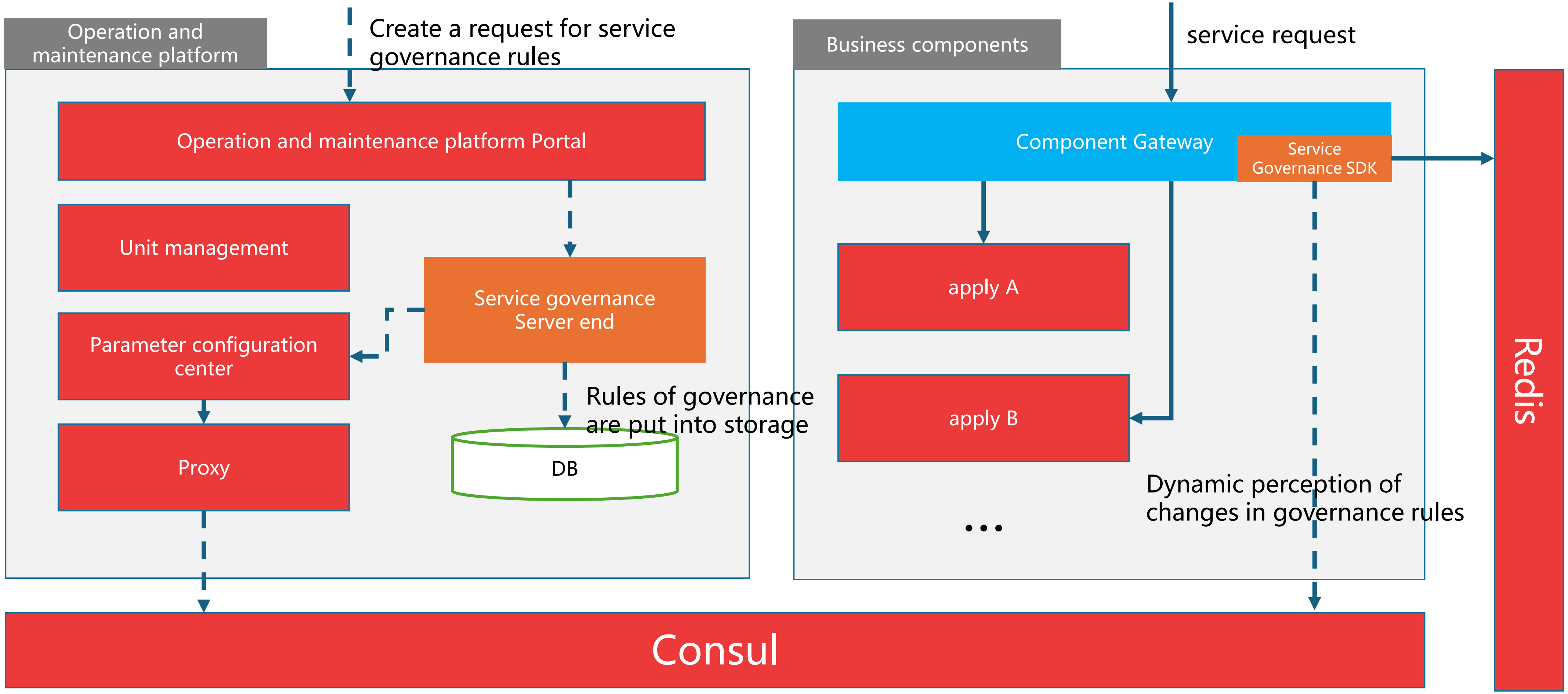
Advantages of microservices architecture

50%
Deployment efficiency improved

99.9%
system availability

70%
Improved resource utilization

Service governance application architecture



Explaination:

- The service governance rules created by the front end request are saved in the database and published to Consul by calling the interface of parameter configuration center;
- The business application system relies on the service governance SDK to realize service authentication, flow limitation, circuit breaking and whitelist verification functions;
- Redis is used to store information such as tokens used to limit traffic.

Evolution Path of Technology Architecture-Microservices and Real-time Capability Construction (Second Stage)

Second Order First Stage: Architecture Advancement

•Add service mesh (Istio/Linkerd)

•Enable full-link tracking (Jaeger/SkyWalking)

•Establish Chaos Engineering Platform to Improve System Resilience

•Complete database read-write separation and sharding

2nd 3rd: Experience Upgrade

•Achieve hyper-personalized services based on real-time customer profiles

•Build an event-driven architecture to support real-time customer interaction

•Achieve seamless omnichannel experience

•Introducing VR/AR Technology to Pilot New Interactive

Second Order Second Stage: Data Intelligence

•Build a unified data fabric

•Deploy a real-time decision engine to support Next Best Offer

•Implement a machine learning platform to support model training and deployment

•Establish real-time customer behavior analysis capability

Stage 4 of the second order: ecological expansion

•Establish an open banking platform to support the API economy

•Unified access and management of third-party services

•Build a developer portal to promote ecosystem collaboration

•Complete cross-border digital banking capability expansion

Customer experience upgrade: Seamless omnichannel experience + Super personalized service

Multi-channel service integration

- **Interface Optimization:** Clear functional zoning and visual guidance with key high-frequency features displayed at the top of the homepage
- **Smart Search:** Supports global search to quickly find feature entries or transaction records
- **Interaction Design:** Simplify and streamline workflows, reduce steps, and improve efficiency
- **Channel integration:** Seamless switching between online and offline services, with real-time synchronization of business processing progress

Dynamic User Profile and Precision Recommendation

- **Intelligent Recommendation Engine:** Differentiated Recommendations Based on Transaction Behavior, Asset Allocation Preferences, and Life Cycle Stages
- **Personalized interface customization:** Users can customize home page modules to achieve a "tailor-made" interface layout
- **Scenario-based active service:** Event-driven reminders with personalized prompts based on user behavior
- **Financial Plan Customization:** Providing Combination Suggestions for Life Stage Goals such as Home Purchase, Car Purchase and Education

type of service	Current capabilities	objective ability	Enhance
product recommendation	Rule-based	**AI algorithm recommendation**	Conversion rate increased by 40%
Customize interface	Fixed Layout	**A thousand faces for a thousand people**	User satisfaction increased by 35%
active service	passive correspondence	**Scenario Trigger**	User activity increased by 50%
wealth management	standard product	**Intelligent Investment Advisor**	AUM increased by 30%

Intelligent Recommendation Engine: Building Personalized Recommendation Capabilities for "Thousand Faces for Thousand People"

The foundation: Data and platform construction

- **Unified Data Platform:** Consolidates all user behavior data across and beyond the APP
- **User Label System:** Constructing Structured Labels with Static and Dynamic Labels
- **Real-time data processing:** ShenCe data collection captures user actions in real time
- **LBS tag:** Recommend nearby life services based on your location

Recommended Algorithm Engine Technology Stack

coordinated filtering | Content Recommendations | association rule

deep learning model

Front-end Presentation: Scenario-based and Intelligent Experience

Customize homepage
The modular homepage allows users to manually adjust it like building blocks

Smart Search
When entering keywords, prioritize displaying the features users are most likely to need.

Timely recommendations
Scene trigger, provides timely product recommendations in the process

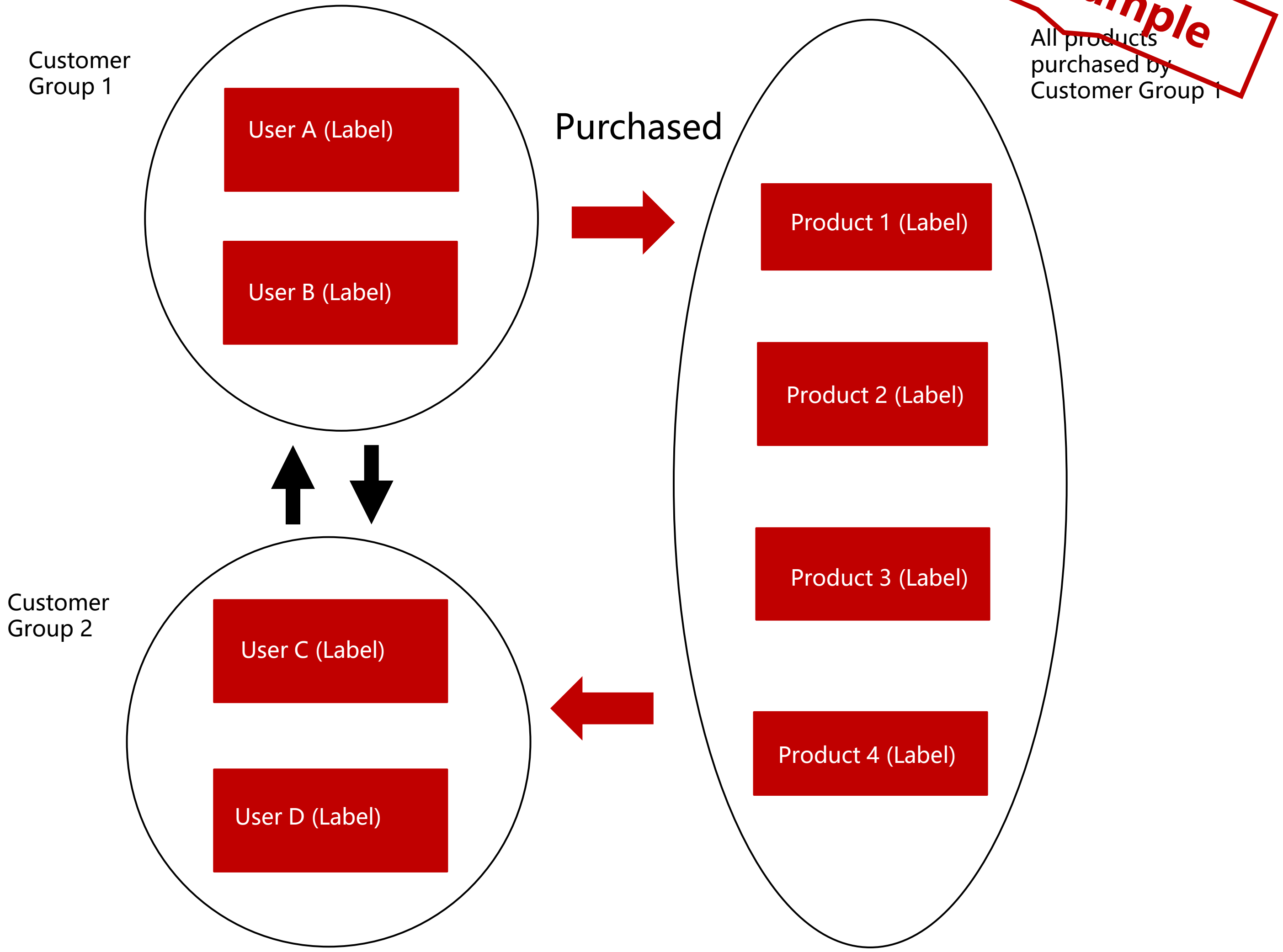
Customer Group Recommendations

Implementation approach: Users are segmented into customer groups based on predefined categories (e.g., user tags). The system first identifies the users current group and assigns tags to it, then searches for similar groups. It then retrieves all product tags purchased by members of these similar groups, merges them, calculates totals, and ranks the results to identify the top 5 product tags. These top 5 tags are then used to match products in the product tag database that contain them. Products that have been purchased are excluded from recommendations.

Algorithm: Based on user-CF (User-Centered Recommendation)

Algorithm formula:

$$P(u,i) = \sum_b \frac{n_{u,b}}{\log(1+n_b^{(u)})} n_{b,i}$$



Customer Group Recommendations (Groups 1 and 2 are similar)

A Thousand Faces Labeling Portrait System

UOB iSoftStone Digital

Sample

label image system

Responsible for managing and applying customer tags related to wealth management services

Core features: Multi-mode self-creation, Visualized tag system, Historical tag search, and Refined tag management

User Tag System Sensors Personas

Create label
- Create a label for the visualization interface
- Import external labels
- SQL Label

View tags
- Label system visualization
- View tag results in real time
- Label history

Tag Application
- Customer Screening
  - Filter by label
  - Filter by user attributes
  - Filter by user behavior
  - Upload the list to create a customer group
- Application of screening results
  - Create customer profile
  - Expand analysis dimensions
  - Export List

Manage tags
- Edit Label
- Turn label on/off
- Tag management permissions

data base platform Sensors Data Platform

DA	User Tag Engine	Permissions and Accounts	data mining
Tracking point management	metadata management	data modeling	system configuration

Thousand Faces for Thousand People Technology Base

UOB iSoftStone Digital

Smart Frontend

Android | iOS | small routine | H5 | IoT

scene

Custom skin | Customize habits | Message customization | Personalized Recommendations | Growth System

product

Mobile analytics | Mobile Push | Move and sync | Smart Placement | Intelligent Prediction

Geofencing | AI originality | Terminal Intelligence | Application Center | AB test

Benefits Center

Data Middle Platform (Traditional Platform)

User Segmentation | Label Center | Smart Recommendations | Rule Management

data mart | real-time computation | Offline calculation | data presentation

Data Source Management | Bloodline management

DA

Client

Page tracking | Business event tracking | LBS

Field data

Business data | Rights data

Mobile banking near real-time data hub logical architecture-scenario sample

UOB iSoftStone Digital

Sample

Scenario: Customers with a monthly balance of more than 50,000 yuan log in to the mobile banking and complete one transaction (pay water fee) with a and get a 5 yuan wechat red envelope

Channel/terminal

Mobile Banking

iOS/Android app/Mini Program/H5
Biometrics/security keyboard

1. Complete the transaction

External ecology

Cooperate with merchants and open bank API

Access security

WAF/CDN, mTLS, device fingerprint,

API gateway / BFF

OAuth2, JWT rate limiting/circuit breaking/grey flow, GraphQL/REST

Dialogue/risk control upfront

Login, risk control challenges, risk scoring

Domain microservices

Account services

Accounts/ balances

Transaction/payment

Transfer/payment/clearing

2. Trade off CDC

Kasumbang

Cards/bills/installments

Customer / Portrait

KYC/Label/Preference

marketing campaign

Customer base & Rules & Products

Center of equity

Points/ cards/certificates/Available Game Plays

3 CDC-> Incident

Real-time processing/reading of models

Flow processing

Flink/project/aggregate/deaduplicate

Risk control/anti-money laundering

Real-time features + rules/models

The CQRS reads the library

Redis / Impala/ DB /

4. Flow processing and query

Real-time marketing

Real-time marketing

Decision engine

Rules/online reasoning, A/B

Reach out to the editor

Push/In-site message/SMS/WeChat; frequency control/deletion/return

Data synchronization and receipt

Equity change, portrait update

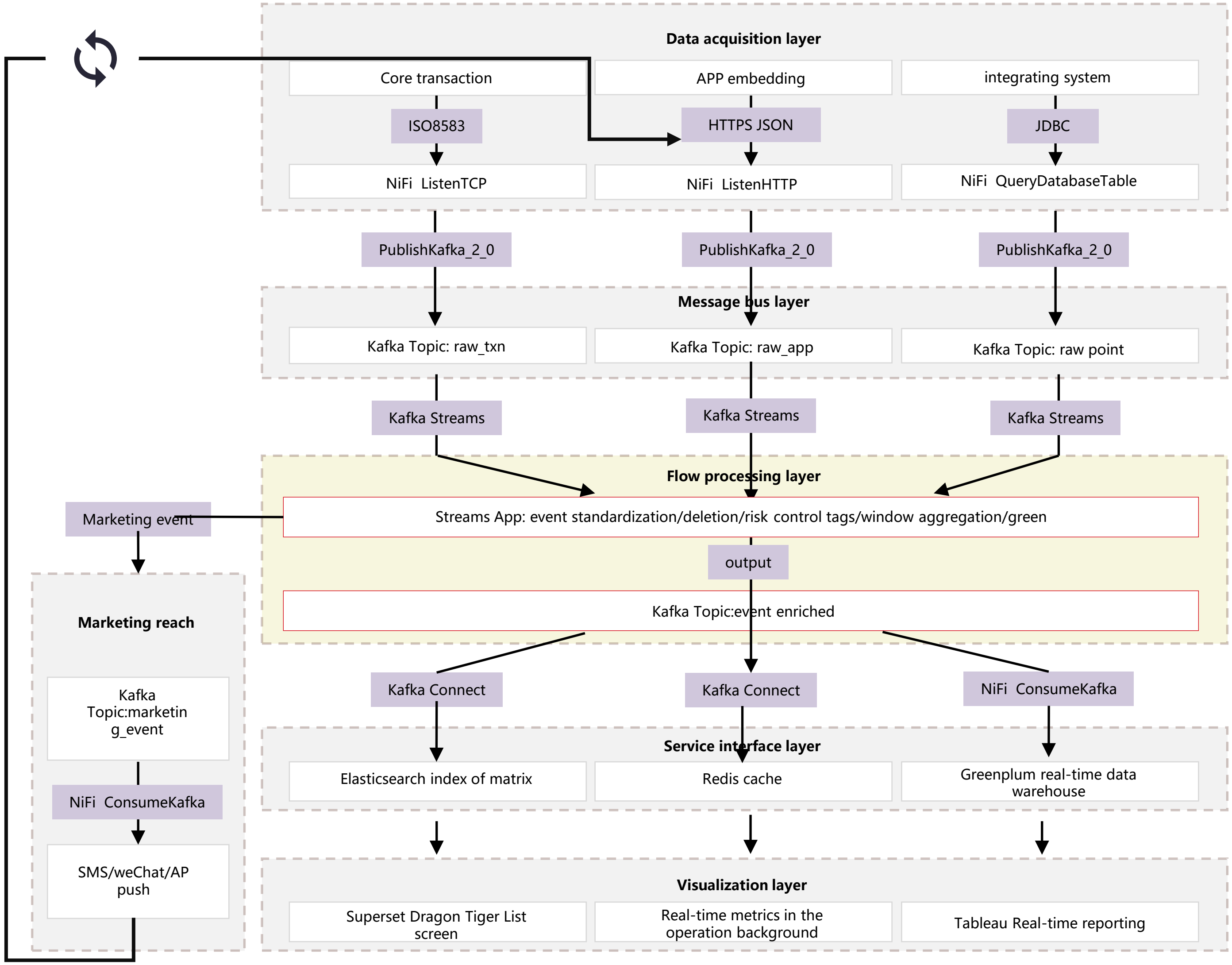
Event streaming platform
CDC event collection and distribution Kafka + Arvo)

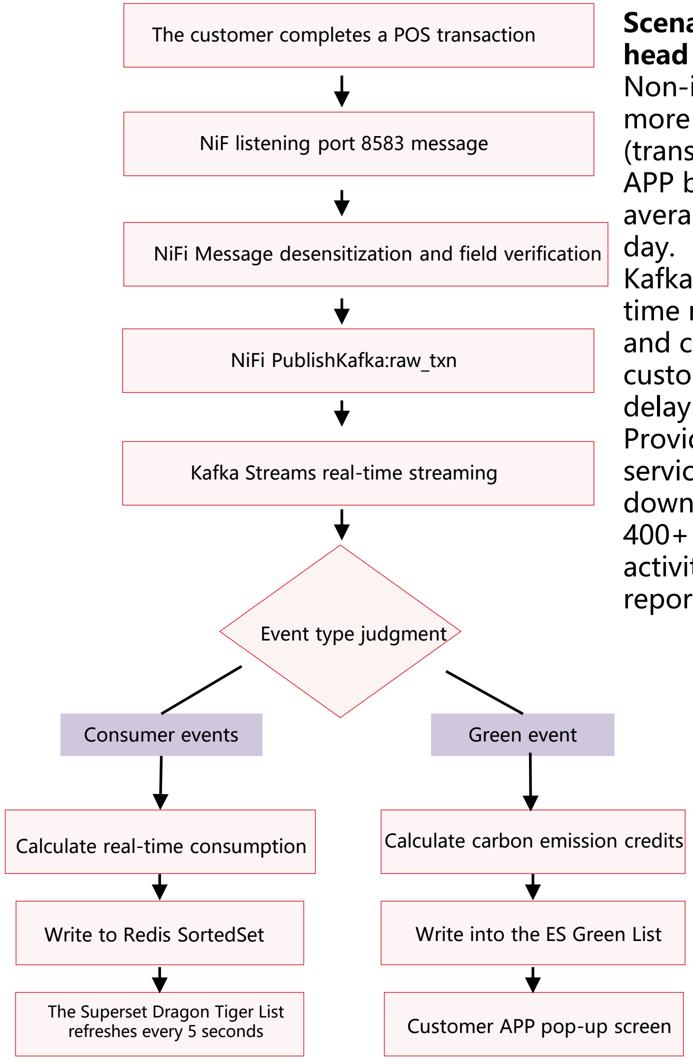
Topics: transaction.eve nts, account.events, customer.events, marketing.events Distributes subscriptions to support multiple channels downstream sinks and triggers rule judgments

Mobile banking transaction closed loop: terminal → API gateway → transaction service landing-CDC capture → Kafka event → flow processing update balance projection/risk control-> CQRS read database to return new balance → notification center receipt
Real-time marketing closed-loop: transaction event → feature update/audience circle selection (Feature Store) decision engine (rules/models/A-B) → reach orchestration-> reach and conversion effect event feedback

Real-time data hub logic architecture and business process sample (NIFI+Kafka)

示例





Scenario target of a domestic head bank

Non-intrusive real-time access to more than 70 upstream systems (transactions, payments, points, APP behavior, etc.), with an average of 150 million data per day.

Kafka Streams can complete real-time recognition, aggregation and completion of more than 200 customer behavior events with a delay of ≤ 80 ms.

Provides high concurrency query service (TPS>10 000) to 20+ downstream systems, supporting 400+ real-time marketing activities and 80+ real-time reports.

Precision marketing architecture

Comprehensive Solution: Guided by the goal of expanding wealth management scale, we will implement data aggregation, build a middle platform, and empower dat Sample s.

Data Application Empowerment

- Objective-driven: Objective and Scientific Quantification of Business Status and Development
- Analysis Drive: Deep Analysis, Opportunity Discovery and Effect Evaluation
- Intelligent Driving:Realizing the Intelligence of Online Platform Based on Data

Data Middle Platform Construction

- Flexible, real-time self-service reporting and analytical query capabilities
- Visualize customer tag portrait production, integration and management
- Intelligent and Automated Customer Segmentation and Differentiated Operations
- Personalized products and content recommendation and display

data infrastructure

- Complete end-to-end capabilities for data collection, validation, and integration
- Integrate underlying data to build a unified user ID system
- Enhanced data management and transformation capabilities for service applications

application scenarios

Data-driven closed-loop management process of financial services

Operation status diagnosis | Establishment of the operational system | Marketing Plan Development | Marketing Execution and Evaluation | Optimize operational strategies

Data Platform

Data Analysis Platform

Custom Overview	Funnel analysis	distribution analysis
event analysis	user path analysis	Attribution analysis
Operation status diagnosis	Operation status diagnosis	Operation status diagnosis

Label Management Platform

Tag Management	Create tag
Import Labels	Label Analysis
Image Insights	Label Output

Smart Operations Platform

Create an operation plan	Automated Process Management
audience insight	planned effect evaluation
Multiple touch channels	Global Configuration Management

data management

data model / **data storage**

text processing | format conversion | Data cleaning | Polymerization | ID-Mapping

metadata management | data quality management | assignment dispatch

DA

Client acquisition: APP | H5 | web | small routine

Server-side collection: server | data base | Import historical data | ...

Data integration across other business systems: data base | data base | ...

Customer touchpoints: Online/Offline Customer Acquisition → Activate → Page navigation & feature clicks → register and login → Browse products/content → Favorites/Add to Cart → deals are done → Repurchase → be sunk in sleep → run off

Marketing support based on AI data mining

Sample

ODS

data handling

data processing

- index processing
- SQL create
- Data cleaning
- characteristic derivative
- data lineage
- Indicators Management
- Analysis of the Second Generation of Central Bank Credit Information
- Third-party data source parsing

data API

- Real-time data processing API
- Batch dataset API
- Download the real-time SDK

DA

data exploration

- Univariate statistical analysis
- multivariate statistical analysis
- Customer Exploration
- SQL Unified Exploration
- Python exploration
- SAS exploration

Mis reporting system

- BI report forms
- Big Data Screen
- ...

model

model analysis

automated modeling expert modeling	univariate attribution multivariate attribution
AutoML technology custom algorithm	characteristic derivative Variable filter

Model Management

Register or publish the model	Champion Challenger
Model version management	Model Comparison

model monitoring

model monitoring	Variable monitoring
System Monitoring	call monitoring

Model Application

real-time reasoning	Batch Reasoning
flow reasoning	...

policy

Strategy Analysis

Smart Strategy	Expert Strategy
Strategy Comparison	Smart Customer Group

tactical management

Policy Release	Policy version
policy evaluation	AB test

Policy monitoring

Policy Effectiveness Evaluation	Policy monitoring
intertemporal verification	Variable monitoring

Policy Application

decision engine	Offline deployment
Online Reasoning	...

Data modeling process

Data Application

- **Filter premium customers**
- **Reduce operating costs**
- **Early intervention**
- **decision support**

mathematical modeling

- model training
  - feature screening
  - model tuning
- model testing
  - intertemporal verification
- model rule extraction
  - client portrait

Define the target variable, select all features, use default parameters, and build a preliminary model

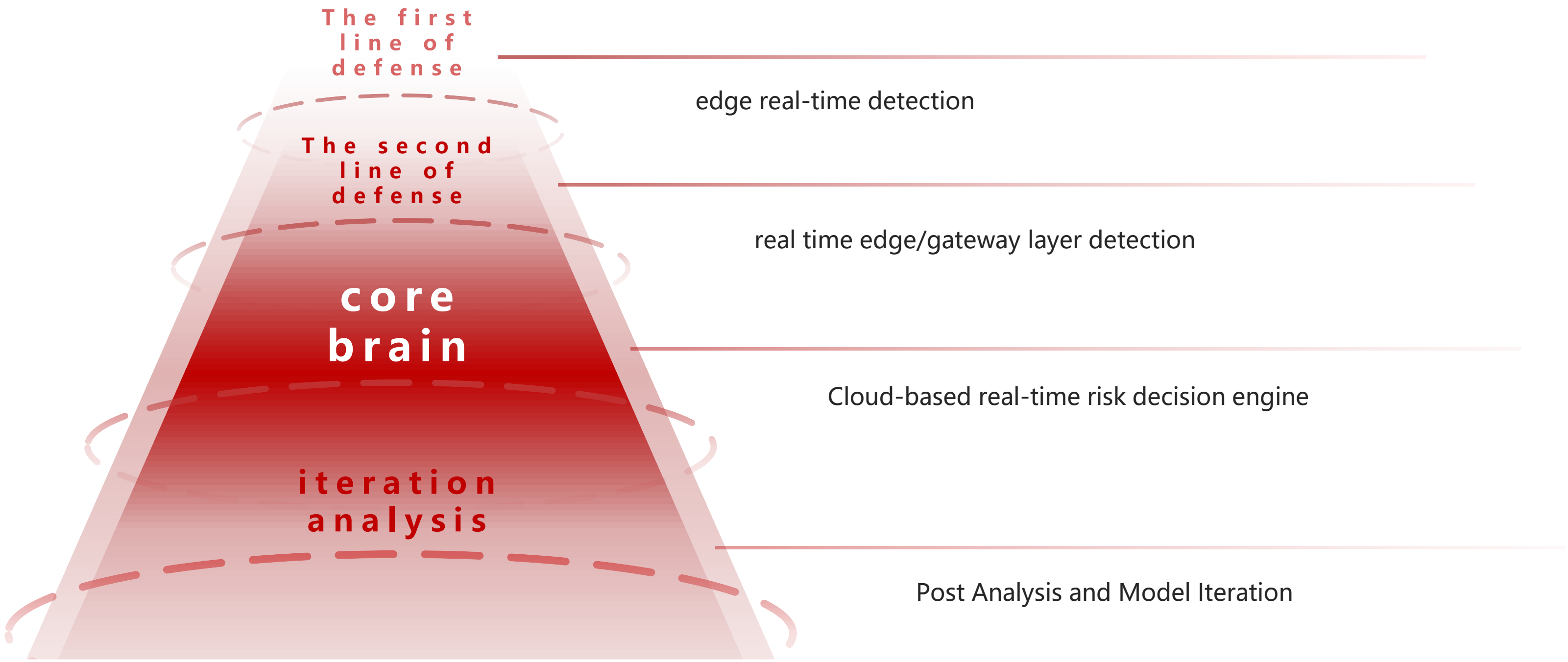
data handling

master table design → Data configuration → Data cleaning → data generation

data generation

- loan transaction information
- Certificate of Deposit Information
- customer information
- Customer Market Information
- off balance sheet transaction information
- on-product information

Real-time fraud detection system: a hierarchical real-time detection architecture



Key technology implementation

rules and models collaboration

- **Hard rule:** For high certainty risks, block directly
- **Machine Learning Model:** Processing Complex and Hidden Correlation Risks
- **Expert Rule:** Rapid Response to New Fraud Model
- **System encryption:** Encrypt and secure the application software before deployment

business process integration

- **Preventive measures:** Device environment detection during APP launch
- **Real-time intervention during the process:** enhanced verification triggered by key nodes
- **Post-incident handling:** automatically stop suspicious transactions and temporarily freeze high-risk accounts

Real-time Customer Behavior Analysis-Precisely Identify Optimal Touchpoints to Drive Personalized Interactions

Data acquisition and analysis platform

front-end data acquisition

- SDK event tracking integration
- full link user behavior tracking
- Real-time data reporting

backend analysis modeling

- big data analysis modeling
- user behavior path analysis
- conversion funnel analysis

Advantages of microservices architecture

path behavior analysis User workflow optimization	**conversion funnel analysis** Key node conversion rate increase	**user segmentation analysis** Refined operation strategy
retained analysis User activity has increased	**thermogram analysis** Interaction Optimization	**real-time event analysis** Instant marketing trigger

Evolution Path of Technology Architecture-Future Bank(3rd Stage)

The "Next-Generation Mobile Banking" project has developed an AI-powered application.

This AI-powered personal mobile banking app leverages large language model technology to deliver personalized greetings, proactive services, AI-driven Q&A, and streamlined transactions, significantly enhancing user experience and service efficiency.

Through big data analytics and user profiling, the system automatically identifies user needs and delivers tailored financial products and promotional offers. Additionally, it provides convenient solutions for bill management, credit card repayments, and fund transfers, establishing a new banking service paradigm for the AI-driven productivity era.

Evolution Path of Technology Architecture-Future Bank-Comprehensive Intelligence

AI-driven Personalized Financial Experience

Future mobile banking will leverage AI technology to evolve from personalized for each individual to customized for all users, delivering truly personalized financial services.

Smart Financial Assistant

Based on user transaction data, behavior patterns, and risk preferences, we provide automated financial advice, spending analysis, and financial planning services.

Voice and natural interaction

A natural language processing-enabled voice assistant that understands complex financial needs and performs operations such as transfers, inquiries, and investments.

predictive financial services

Machine learning is used to predict users financial needs and provide recommendations of credit, investment or insurance products at the right time.

intelligent risk prevention and control

Utilize AI models to monitor transaction risks in real time, enabling millisecond-level fraud detection and dynamic identity verification to ensure fund security.

By 2028, over 70% of mobile banking transactions are expected to be handled through AI assistants, with human agents focusing on complex and special cases.

Technology Architecture Evolution Path-Future Bank-Intuitive Financial Experience

UOB iSoftStone Digital

Intelligent Financial Services Hidden in the Scene

In the future, mobile banking will gradually become invisible, with financial services seamlessly integrated into daily life like water and electricity, allowing users to access them without actively searching.

now

Scan to pay, fingerprint or face recognition

2027+

behavior recognition, environment perception payment

2030+

Brain-computer interface, holographic projection interaction

insensitive financial application scenarios

- Smart car: Automatic parking fee deduction and automatic fuel payment
- Smart Home: Automatic Payment of Water, Electricity and Gas, Automatic Insurance for Appliance Repair
- Health Wearables: Automatic Settlement of Medical Expenses, Dynamic Pricing of Health Insurance
- Business scenario: Automatic promotion push upon entry and automatic checkout upon exit

The core philosophy of Incept Financial is service finds people rather than people find service, which significantly enhances the convenience of financial services and user satisfaction.

Technology Architecture Evolution Path to Future Bank

UOB iSoftStone Digital

Generation Z Consumer Growth: Future 10-Year Trends in Consumption Momentum and Quality

Customer growth: study | work | manage money matters | house-purchase | purchase a car | marry | to have a baby | consume | education | medical treatment

Five Key Elements of Bank Metaverse: Virtual & Real | Connect people, events, and things in real time | digital means of production | Business transcends time and space | Close the relationship with consumers

Human being

- Digital identity
- financial feature
- Social attributes

individual
Associatio n
family
relatives and friends

On-line

- Service is available in real time
- immersive interactive experience
- Digital person guidance

virtual space
+
present world

Offline

- Offline scenarios linked to digital human guides

Money

- Digital Collection
- financial products
- physical commodity

immerse
AUM
social intercourse

High frequency/high viscosity/immersive/social scene building

Process transactions in real time to strengthen consumer relationships

Business processing scenarios	Credit Card Installment-E-commerce Scenario
Financial management purchase scenario	Marketing Campaign Scenario

Virtual space + digital human guidance/live streaming, combined with business scenarios

Boost customer growth and social engagement to drive user expansion

Energy Exchange Collection/Benefits Scenario	Collection Exchange/Purchase/Transfer
User Space and Collection Gallery	User journey collection and growth review

Virtual Space + Digital Collectibles, Combining Points System + Event Operation Scenarios

Virtual reality integration for bidirectional traffic diversion

Offline service	Offline benefit redemption
Offline marketing interaction	Offline community events

Virtual Reality + Digital Human Guidance, Combined with Offline Business Scenarios

Technology Architecture Evolution Path to Future Bank Operation Scenario

Driving Traffic

Scene 1: Welcome Area	Business processing, digital human guidance, or live streaming
Scene 2: Credit Card Hall	Credit Card Car and Commodity Installment, Product Introduction
Scenario 3: Wealth Hall	Financial products, digital human guidance and introduction
Scene 4: Live Studio	Supports live streaming, digital human live streaming, or real face swapping
Scenario 5: Personal Center	User digital assets, digital libraries, and social attributes

Attraction

Scenario 1: Offline Welcome	Business processing, virtual reality, digital human guidance
Scenario 2: Offline Marketing	Online benefits redeemed offline, offline marketing drives online traffic

Technology Architecture Evolution Path to Future Bank Digital Virtual Human Application Scenarios

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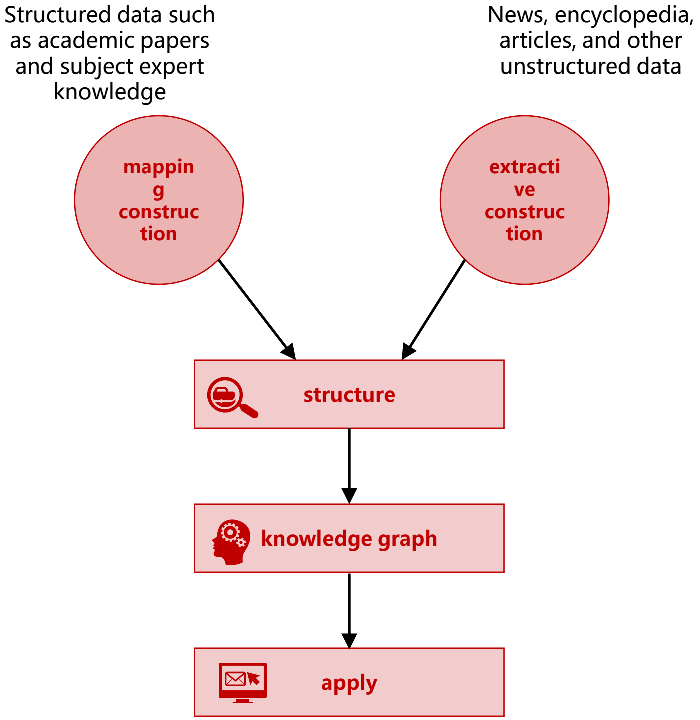
User Knowledge Base Construction



Semantic search, intelligent recommendation, Q&A system, path analysis, relationship mining, professor profile, authority profile, visual analysis, decision-making model, expert system...

front-end data acquisition

Supported scenarios: digital human Q&A, digital human live streaming, and live streaming face swapping.

Technology Architecture Evolution Path to Future Bank Business Service Scenario

Example: Guest House
Similar to setting up flagship offline store scenarios

Scenario value:
The front-end immersive experience enables hosting online virtual space events of any scale. Digital avatars establish brand identity, offering real-time online greetings and explanations about marketing campaigns or trending products. By replacing human staff, it reduces labor costs, fosters interactive warmth, bridges the gap with users, and elevates brand value.

iSoftStone Professional Service for Mobile Banking Development

UOB iSoftStone Digital

Core Competencies and Strengths

Rich Mobile Banking Experience

- Strong product and roadmap design experience
- Served more than 50 large financial institutions.
- Delivered > 200 Mobile banking projects
- Familiar with the banks core system, channel system, and risk control system, data platform

Technical implementation capability

- Microservice Architecture Design and Implementation Expert
- Cloud Native Technology in Depth
- Big Data and AI Technology Integration
- DevOps and Automation Toolchain

Team resource support

- over 1000+ professionals in the mobile banking domain of financial technology
- Combination of financial business experts and technical experts
- Agile Development and Hybrid Model of Waterfall Development
- 24/7 operation and maintenance support

Project management capability

- CMMI Level 5 Certification Delivery System
- financial grade quality management standard
- Risk Early Warning and Response Mechanism
- Transparent project progress management

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